

L2.6a Optimization

Recall: L2.5 Summary of Curve Sketching

Name _____

Solo – 9 minutes

1. For $f(x) = \frac{x^2 - 4}{x^2 - 1}$, name the vertical asymptotes, the horizontal asymptote, and the intercepts.
Is f even, odd, or neither?

2. Find the absolute Max and the absolute min of $g(x) = x^3 - 6x^2 + 5$ on $[-1, 5]$.

3. A rectangle has sides x and y satisfying $2x + y = 40$, with $x > 0$ and $y > 0$.

(a) What values can x take?

(b) Now suppose neither side may be shorter than 5 ft. Now what values can x take?